

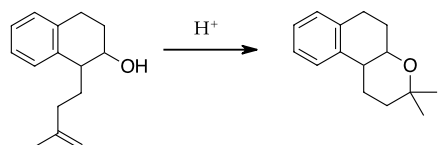
Mechanism Questions #1

Barry Linkletter

Explain the reaction mechanisms for the transformations below. Draw out each step and show electron movement using curved arrows. Identify the likely rate-determining step and identify the driving force for each reaction.

Each reaction is a single step in a multistep synthesis described in the referenced literature. For extra experience, access each paper and describe every step in the synthesis that is presented.

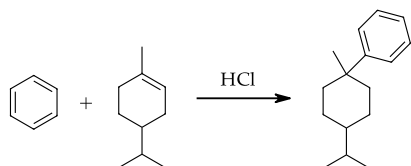
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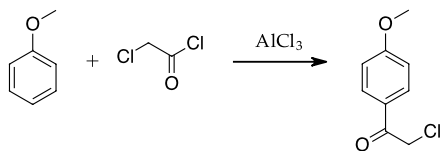
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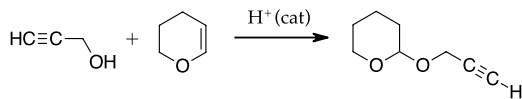
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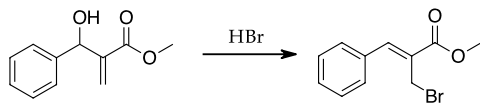
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5.



6.



This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in ChemDoodle.

"Cyclizations wherein an epoxide acts as the source of initiation and termination steps. Evidence for an early transition state in biomimetic epoxide cyclizations." Stephen K. Taylor, Scott A. May, Jeffrey A. Hopkins, *Tetrahedron Letters*, 1993, 34, 1283-1286. [https://doi.org/10.1016/S0040-4039\(00\)91775-1](https://doi.org/10.1016/S0040-4039(00)91775-1). Available in UPEI Library stacks

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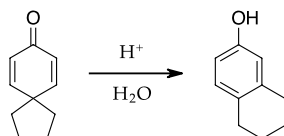
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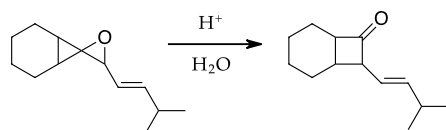
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7.



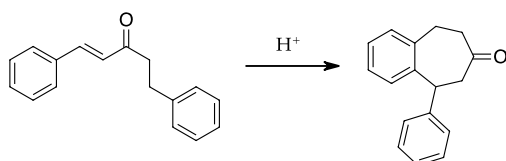
"Ein Beitrag zur Dienon-Phenol-Umlagerung." Bernd Hagenbruch, Siegfried Hünig, *Chem. Ber.*, **1983**, *116*, 3884-3894. <https://doi.org/10.1002/cber.19831161212>

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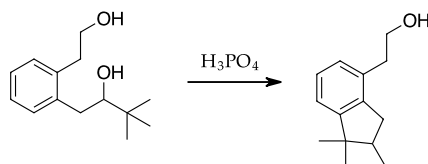
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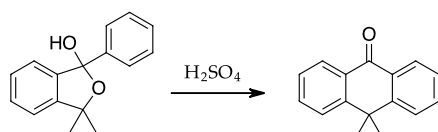
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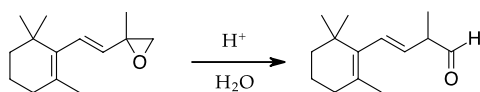
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"An expedient route to a versatile intermediate for the stereoselective synthesis of all-trans-retinoic Acid and beta-carotene." James H. Babler, Scott A. Schlidt, *Tetrahedron Letters*, **1992**, *33*, 7697-7700. [https://doi.org/10.1016/0040-4039\(93\)88020-J](https://doi.org/10.1016/0040-4039(93)88020-J). Available via UPEI Library stacks.

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will have...

- reviewed organic chemistry mechanisms involving addition reactions or acidic conditions.